

PCIe 7.0 Controller IP

Technical Specification

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Top module	<code>pcie_controller_top</code>
Technology	Process-independent RTL
Classification	Confidential

Note: This document describes the `pcie_controller_top` IP as an engineering deliverable. It is not a PCI-SIG compliance certification. All compliance claims are limited to the evidence in `docs/compliance_matrix.md`.

0 Revision History

Version	Date	Author	Changes
0.1.0	2026-05-17	Design Team	Initial commit: full RTL stack, 10-test directed TB
0.2.0	2026-05-21	Design Team	Added: SVA suites, UVM PIPE agent, error injection, PM RTL, CD

Contents

1	Introduction	2
1.1	Purpose	2
1.2	Scope and Limitations	2
1.3	Referenced Documents	2
2	Feature Overview	2
2.1	Key Features	3
2.2	Top-Level Parameters	3
3	Architecture	3
3.1	Block Diagram	4
3.2	Layer Responsibilities	4
4	Port List	5
4.1	Clock and Reset	5
4.2	PIPE Interface	5
4.3	AXI4 Subordinate Interface	6
4.4	AXI4 Manager Interface	6
4.5	DMA Interface	6
4.6	Interrupt Interface	6
4.7	Status and Debug	6
5	Protocol Description	6
5.1	Link Training	6
5.2	Data Link Layer	7
5.3	Flow Control	7
5.4	TLP Types Supported	7
6	Configuration Space Register Map	7
7	Clock and Reset Architecture	8
7.1	Clock Domains	8
7.2	CDC Crossings Summary	9
7.3	Reset Sequence	9
8	Power Management	9
8.1	Power States	9
8.2	UPF Power Domains	9
9	Verification Collateral	9
9.1	Directed Testbench (iverilog)	10
9.2	UVM Testbench	10
9.3	SVA Modules	10

10 Delivery Checklist	11
Acceptance Criteria	11

1 Introduction

1.1 Purpose

This specification defines the functional and interface contract for the `pcie_controller_top` PCIe 7.0 Controller IP. It covers:

- Feature set and parameterization
- Port list and interface definitions
- Protocol layer descriptions (Physical, Data Link, Transaction)
- Configuration space register map
- Power management architecture
- Clock domain and reset structure
- Verification and delivery collateral

1.2 Scope and Limitations

This IP is a simulation-verified RTL model for educational, portfolio, and prototype integration use. The following are **outside scope**:

- PCI-SIG compliance certification or logo program qualification
- PHY electrical layer (represented by PIPE-level BFM)
- FLIT-mode framing (planned for v0.3.0)
- Ordering rule enforcement (planned for v0.3.0)

1.3 Referenced Documents

Reference	Document
[PCI-SIG-7]	PCI Express Base Specification, Revision 7.0
[PIPE-6]	PCI Express PHY Interface for the PCI Express Architecture, v6.0
[AXI-AMBA]	AMBA AXI and ACE Protocol Specification, v2.0
[IEEE-1801]	IEEE Standard for Design and Verification of Low-Power, Energy-Aware Electronic Systems
[IEEE-1685]	IEEE Standard for IP-XACT
[IEEE-1800]	IEEE Standard for SystemVerilog

2 Feature Overview

2.1 Key Features

Feature	Description
PCIe Generation	Gen1 (2.5 GT/s) through Gen7 (128 GT/s), parameterized
Lane Width	×1, ×2, ×4, ×8, ×16
Device Role	Endpoint (EP), Root Port (RP), Dual Mode (DM), Switch Port (SW)
AXI Interface	AXI4 Subordinate (inbound) + AXI4 Manager (completions), 256-bit
DMA Engine	4-channel H2D/D2H, MRRS/MPS-bounded chunking, 256-tag pool
MSI / MSI-X	Up to 32 MSI vectors; up to 2048 MSI-X vectors
AER	Advanced Error Reporting: correctable, non-fatal, fatal
Flow Control	Credit-based FC: Posted, Non-Posted, Completion (infinite support)
DLL	12-bit sequence numbers, LCRC-32, 4096-entry replay buffer
LTSSM	Full 28-state machine per PCIe 7.0 §4.2
Power Management	L0s auto-entry, L1 DLLP handshake, L2 software entry, PME wake
Completion Timeout	Per-tag countdown timers, Range B default (50 ms), correctable error
CDC Synchronizers	2-flop sync, pulse sync, reset sync, async FIFO (Gray-code)
Config Space	4 KB Type-0/Type-1, PCIe Cap, PM Cap, MSI, MSI-X, AER

2.2 Top-Level Parameters

Parameter	Type	Default	Description
DEVICE_ROLE	pcie_role_e	ROLE_EP	EP / RP / DM / SW
MAX_GEN	pcie_gen_e	PCIE_GEN7	Maximum PCIe generation
NUM_LANES	int	16	×1 to ×16
DATA_W	int	256	Internal / AXI data width (bits)
ADDR_W	int	64	AXI address width (bits)
AXI_ID_W	int	8	AXI ID width (bits)
PIPE_W	int	32	PIPE data width per lane (bits)
VENDOR_ID	logic[15:0]	16'hCAFE	PCIe Vendor ID
DEVICE_ID	logic[15:0]	16'h0001	PCIe Device ID
BAR0_APERTURE	logic[7:0]	24	BAR0 size: 2 ⁿ bytes
EN_MSI	bit	1	Enable MSI capability
EN_MSIX	bit	1	Enable MSI-X capability
EN_AER	bit	1	Enable AER extended capability
EN_ECRC	bit	1	Enable ECRC generation/checking
EN_DMA	bit	1	Enable DMA engine
DMA_CHANNELS	int	4	Concurrent DMA channels

3 Architecture

3.1 Block Diagram

Figure 1 shows the top-level functional block diagram.

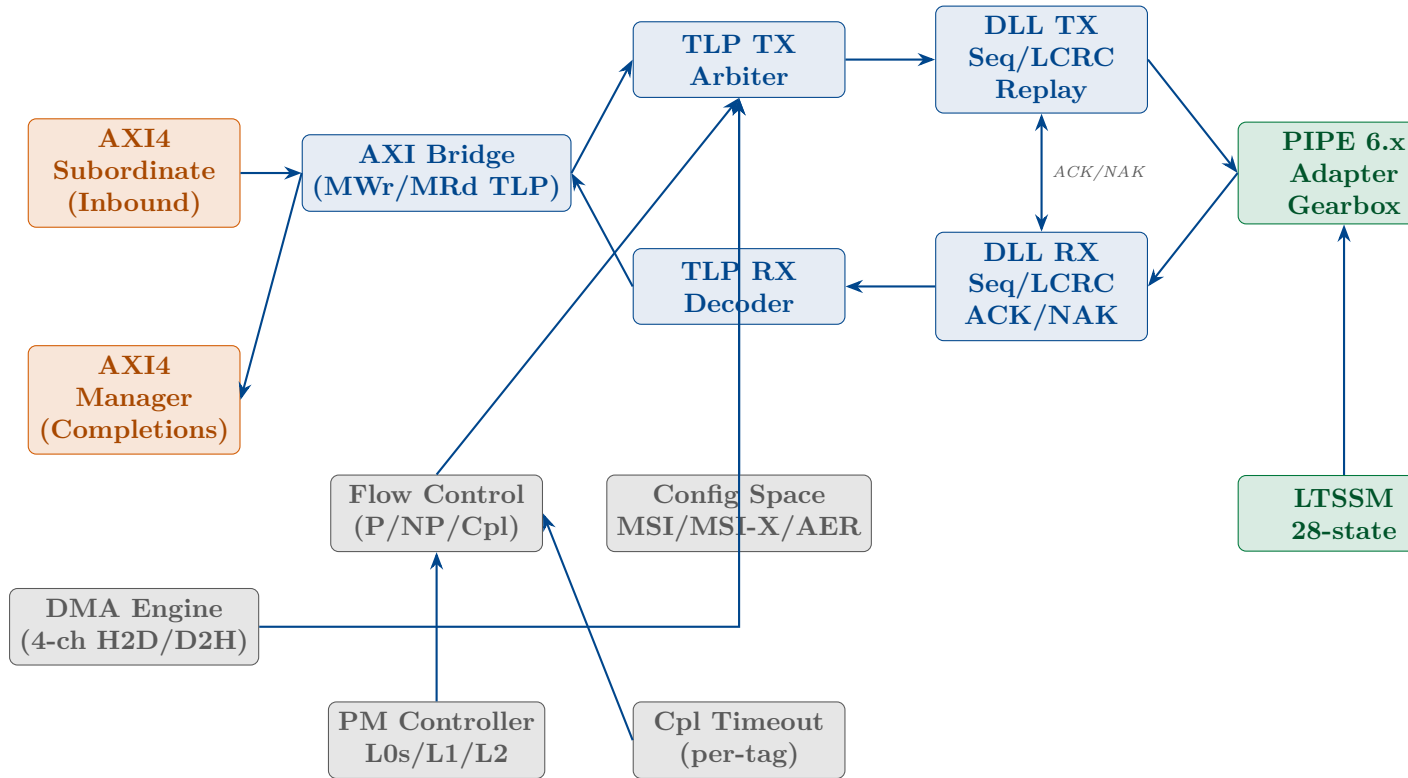


Figure 1: PCIe 7.0 Controller — Top-Level Functional Block Diagram

3.2 Layer Responsibilities

Physical Layer

LTSSM manages the 28-state link training machine. The PIPE 6.x adapter implements the TX gearbox (256-bit $\rightarrow N \times \text{PIPE_W}$), K-symbol insertion (STP/SDP/END_/EDB), and RX assembly. CDC synchronizers (`pcie_cdc_sync.sv`) and async FIFOs (`pcie_async_fifo.sv`) handle the `core_clk/pipe_clk` cross-ing.

Data Link Layer

TX assigns 12-bit sequence numbers and appends CRC-32 LCRC. A 4096-entry replay buffer holds outstanding TLPs. RX validates sequence and CRC, generates ACK/NAK DLLPs, and decodes incoming DLLPs (FC Init, FC Update, PM).

Transaction Layer

TX arbiter round-robins four input ports (AXI bridge Posted, AXI bridge Non-Posted, DMA, Config Space), gated by Flow Control credits. RX decodes the 3DW/4DW header, routes to the appropriate sink (AXI bridge, config space, DMA, completion buffer), and reports poisoned/malformed TLP errors.

AXI Bridge

Maps AXI4 write transactions to Memory Write TLPs (MWr32/MWr64) and AXI4 read transactions to Memory Read TLPs (MRd32/MRd64), with a 256-entry tag pool for outstanding reads. Returning CplID are reassembled into AXI R-channel responses.

DMA Engine Issues host-to-device MRd64 TLPs (MRRS-bounded), assembles completions, and writes to local AXI memory. Device-to-host issues MWr64 TLPs (MPS-bounded). Uses tag range 512–767, reserving 0–511 for AXI bridge.

Config Space Implements a 4KB Type-0/Type-1 configuration space with capabilities at fixed offsets: PCIe Cap (0x40), PM Cap (0x80), MSI Cap (0x90), MSI-X Cap (0xA0, 2048 vectors), AER Extended Cap (0x100). RW1C error registers.

Power Management

`pcie_pm_ctrl` FSM handles L0s auto-entry on idle, L1 DLLP handshake (PM_Enter_L1 / PM_Req_Ack), software-initiated L2, and PME wakeup. Drives DLLP requests to DLL TX and LTSSM power-state requests.

Completion Timeout

`pcie_cpl_timeout` maintains per-tag countdown timers. On expiry (default Range B, 50 ms), pulses `cfg_err_cor` and reports the offending tag.

4 Port List

4.1 Clock and Reset

Port	Dir	Width	Description
<code>core_clk</code>	in	1	Core / application clock
<code>core_rst_n</code>	in	1	Active-low async reset; deassert synchronous to <code>core_clk</code> via <code>pcie_rst_sync</code>
<code>pipe_clk</code>	in	1	PIPE reference clock from PHY
<code>aux_clk</code>	in	1	Always-on auxiliary clock for power management

4.2 PIPE Interface

Port	Dir	Width	Description
<code>pipe_tx_data</code>	out	$N \times \text{PIPE_W}$	TX data (per lane)
<code>pipe_tx_data_k</code>	out	$N \times \text{PIPE_W}/8$	TX K-symbol indicator
<code>pipe_tx_elec_idle</code>	out	N	Electrical idle request
<code>pipe_tx_compliance</code>	out	N	Compliance pattern enable
<code>pipe_tx_deemph</code>	out	N	TX de-emphasis select
<code>pipe_tx_margin</code>	out	$N \times 3$	TX margin control
<code>pipe_tx_swing</code>	out	N	TX voltage swing
<code>pipe_tx_eq_ctrl</code>	out	$N \times 2$	Equalization preset
<code>pipe_rx_data</code>	in	$N \times \text{PIPE_W}$	RX data (per lane)
<code>pipe_rx_data_k</code>	in	$N \times \text{PIPE_W}/8$	RX K-symbol indicator
<code>pipe_rx_valid</code>	in	N	RX data valid
<code>pipe_rx_elec_idle</code>	in	N	Electrical idle detect
<code>pipe_rx_status_valid</code>	in	N	RX status valid
<code>pipe_rx_status</code>	in	$N \times 3$	RX status (detect, speed change, ...)
<code>pipe_power_down</code>	out	4	Power-down control
<code>pipe_reset_n</code>	out	1	PIPE reset (active-low)
<code>pipe_rate</code>	out	4	Speed select (Gen1–Gen7)
<code>pipe_width</code>	out	2	Lane width select
<code>pipe_clk_req_n</code>	in	1	Clock request (active-low)

$N = \text{NUM_LANES}$ parameter.

4.3 AXI4 Subordinate Interface

All signals prefixed **s_axi_**; standard AXI4 write and read channels. Data width = DATA_W, address width = ADDR_W, ID width = AXI_ID_W. AWBURST and ARBURST must be INCR (2'b01).

4.4 AXI4 Manager Interface

All signals prefixed **m_axi_**; driven by PCIe completions toward system memory. Same widths as subordinate interface.

4.5 DMA Interface

Port	Dir	Width	Description
dma_start	in	1	One-cycle pulse: start DMA transfer
dma_src_addr	in	64	Source address (host or device)
dma_dst_addr	in	64	Destination address
dma_length	in	32	Transfer length in bytes
dma_dir	in	1	Direction: 0 = H2D, 1 = D2H
dma_done	out	1	One-cycle pulse: transfer complete
dma_error	out	1	One-cycle pulse: transfer error

4.6 Interrupt Interface

Port	Dir	Width	Description
msi_irq	out	1	MSI interrupt pulse
msi_vector	out	5	MSI vector index (0–31)
msix_irq	out	1	MSI-X interrupt pulse
msix_vector	out	11	MSI-X vector index (0–2047)
intx_assert	in	1	Legacy INTx assertion request

4.7 Status and Debug

Port	Dir	Width	Description
link_up	out	1	Link in L0 state
ltssm_state	out	6	Current LTSSM state
negotiated_gen	out	4	Negotiated PCIe generation
negotiated_width	out	5	Negotiated lane width
cfg_err_cor	out	1	Correctable error detected
cfg_err_nonfatal	out	1	Non-fatal uncorrectable error
cfg_err_fatal	out	1	Fatal uncorrectable error
max_payload_size	out	3	Negotiated MPS (encoded)
max_read_req_size	out	3	Negotiated MRRS (encoded)

5 Protocol Description

5.1 Link Training

The LTSSM implements the full 28-state machine from PCIe 7.0 §4.2:

DETECT → POLLING → CONFIGURATION → RECOVERY → L0

Key transitions are guarded by timeout counters (24-bit). Training sequence detection (TS1/TS2) uses the PIPE `rx_status` port. After `CONFIG_IDLE`, the LTSSM de-asserts `pipe_reset_n`, asserts `link_up`, and enters L0.

5.2 Data Link Layer

- TX** Each TLP receives a 12-bit sequence number (wrapping at 4096) and an appended CRC-32 LCRC. A copy is stored in the 4096-entry replay buffer. The replay timer (default 4096 cycles) triggers retransmission on expiry. The ACK latency timer (default 256 cycles) ensures prompt ACK emission.
- RX** Validates sequence number (no gaps, no duplicates) and CRC. On error, transmits a NAK DLLP. On success, transmits an ACK DLLP and delivers the TLP payload to the Transaction Layer. Decodes FC Init and FC Update DLLPs.

5.3 Flow Control

Credit-based flow control per PCIe 7.0 §2.11:

- Three categories: Posted (P), Non-Posted (NP), Completion (CPL)
- Header credits (12-bit) and Data credits (20-bit, units of 4 DW)
- Infinite credit encoding: all-ones (12'hFFF / 20'hFFFFFF)
- FC Init handshake: INIT1 then INIT2 for each category
- Periodic FC Update: timer-driven, default 200 000 cycles

5.4 TLP Types Supported

TLP Type	Code	Direction
Memory Read 32-bit	MRd32	TX/RX
Memory Read 64-bit	MRd64	TX/RX
Memory Write 32-bit	MWr32	TX/RX
Memory Write 64-bit	MWr64	TX/RX
Config Read Type 0/1	CfgRd0/1	RX
Config Write Type 0/1	CfgWr0/1	RX
Completion	Cpl	TX/RX
Completion + Data	CplD	TX/RX
Message	Msg/MsgD	TX/RX
AtomicOp Request	AtomicOpReq	RX

6 Configuration Space Register Map

Offset	Register Name	Access	Description
0x000	Vendor/Device ID	RO	{DEVICE_ID[15:0], VENDOR_ID[15:0]}
0x004	Command/Status	RW/RO	PCI command (I/O space en, bus master en, ...)
0x008	Revision/Class	RO	{CLASS_CODE[23:0], REVISION_ID[7:0]}

continued on next page

Offset	Register Name	Access	Description
0x00C	Cache Line / Lat.	RW/RO	Cache line size, latency timer, header type
0x010–0x024	BAR0–BAR5	RW	Base Address Registers; BAR0 = 64-bit prefetchable
0x02C	Subsystem IDs	RO	Subsystem vendor / device IDs
0x034	Cap Pointer	RO	0x40 (PCIe Capability)
0x03C	Interrupt Line/Pin	RW/RO	INTx line; pin = INTA
0x040–0x06C	PCIe Capability	RW/RO	Device/Link Capability 1/2, Control, Status
0x040+0x02	PCIe Cap Register	RO	Version = 2, Device/Port type = EP
0x040+0x04	Device Capability	RO	Max payload 256 B supported, extended tags
0x040+0x08	Device Control	RW	MPS, MRRS, relaxed ordering, no-snoop, cpl timeout range
0x040+0x0A	Device Status	RW1C	Correctable/non-fatal/fatal/UR detected
0x040+0x0C	Link Capability	RO	Max link speed Gen7, max width ×16
0x040+0x10	Link Control	RW	ASPM, RCB, link disable, retrain, clock PM
0x040+0x12	Link Status	RO	Current speed, width, training, slot clock
0x080–0x084	PM Capability	RW/RO	PME support D0/D3hot/D3cold; power state
0x090–0x09C	MSI Capability	RW	MSI Enable, 64-bit addr, multiple msg en
0x0A0–0x0A8	MSI-X Capability	RW	MSI-X Enable, table size 2047, table/PBA BIR
0x100–0x11C	AER Extended Cap	RW1C/RW	Correctable/uncorrectable error status/mask/severity
0x100+0x04	Uncorr. Error Stat.	RW1C	DLP error, surprise down, poisoned TLP, FC error, ...
0x100+0x08	Uncorr. Error Mask	RW	Per-bit mask for uncorrectable errors
0x100+0x0C	Uncorr. Severity	RW	0 = non-fatal, 1 = fatal per bit
0x100+0x10	Corr. Error Status	RW1C	Bad TLP, bad DLLP, replay num rollover, replay timeout
0x100+0x14	Corr. Error Mask	RW	Per-bit mask

7 Clock and Reset Architecture

7.1 Clock Domains

Domain	Clock Port	Typical Freq.	Contents
PD_CORE	core_clk	250 MHz	LTSSM, DLL, FC, TL, AXI bridge, DMA, completion timeout
PD_PIPE	pipe_clk	500 MHz (Gen7)	PIPE adapter gearbox
PD_AUX	aux_clk	32 kHz (real) or 100 MHz (sim)	PM controller, retention registers

7.2 CDC Crossings Summary

ID	Crossing	Mitigation	Signals
CDC-001	core → pipe	pcie_sync2 per-bit	pipe_rate, pipe_width, pipe_power_down
CDC-002	pipe → core	pcie_async_fifo	RX data bus
CDC-003	core → aux	pcie_sync2 per-bit	PM control signals
CDC-004	aux → core	pcie_sync_pulse	PME wake pulse
CDC-005	async → *	pcie_rst_sync	core_rst_n deassertion

7.3 Reset Sequence

1. Assert `core_rst_n` low (active). All registers initialize to reset state.
2. Bring up `core_clk` and `aux_clk`; initialize PHY.
3. When PHY asserts `pipe_clk` valid, bring up `pipe_clk`.
4. Hold all clocks stable for ≥ 100 ns.
5. Deassert `core_rst_n` high; synchronized internally by `pcie_rst_sync`.
6. LTSSM starts from DETECT_QUIET automatically.

8 Power Management

8.1 Power States

State	Entry	Exit	Description
L0	Power-on wakeup	/ N/A	Full operation; <code>link_up</code> asserted
L0s	Auto (64+ idle cycles)	Any TLP	Low-latency autonomous power save; no handshake
L1	SW request + DLLP handshake	PME / SW	PM_Enter_L1 / PM_Req_Ack exchange; PIPE powered down
L2	SW via PM register	PME / CLKREQ	Deep sleep; core may power down; aux retained

8.2 UPF Power Domains

Three power domains defined in `rtl/pcie_controller_top.upf`:

- **PD_CORE** (VDD_CORE 1.0 V) — always-on during L0/L0s/L1; may power down in L2
- **PD_PIPE** (VDD_PIPE 1.0 V) — may power down in L1 and L2
- **PD_AUX** (VDD_AUX 0.9 V) — always retained; holds PM FSM state

9 Verification Collateral

9.1 Directed Testbench (iverilog)

Test	Coverage
1. Link Training	LTSSM: DETECT → POLLING → CONFIG → L0
2. Negotiated Params	Gen and width within valid range after link-up
3. Config Space Init	Link stable after config-space initialization
4. MPS/MRRS Fields	Max Payload and Max Read Request Size
5. AXI Write	AXI4 write generates MWr TLP on PIPE TX
6. AXI Read	AXI4 read generates MRd TLP; CplD returns
7. DMA H2D	MRd64 chunked; completions assembled
8. DMA D2H	Local read; MWr64 issued to host
9. MSI Interrupt	INTx assertion triggers MSI IRQ
10. AER Error	No spurious errors in steady-state L0

9.2 UVM Testbench

Component	Description
pcie_axi_agent	Sequencer, driver, monitor on AXI subordinate interface
pcie_pipe_agent	PIPE-level driver (RX stimuli), monitor (TX capture)
pcie_scoreboard	AXI response check; error count
pcie_coverage	AXI direction × response cross coverage
pcie_ltssm_cov	28-state, transition, and Gen×width coverage
pcie_error_regression_test	Bad LCRC, NAK, replay timeout, malformed TLP, poisoned TLP, cpl timeout, FC exhaust

9.3 SVA Modules

Module	Properties
pcie_ltssm_assertions	12 assert + 12 cover: state validity, link_up/L0 consistency, power control, liveness
pcie_dll_assertions	10 assert + 7 cover: backpressure, DLLP type, burst integrity
pcie_fc_assertions	9 assert + 6 cover: credit underflow/overflow, liveness
pcie_tlp_assertions	11 assert + 8 cover: AXI protocol, TLP framing, DMA exclusion
pcie_delivery_assertions	5 assert: TLP X-check, DMA exclusion, backpressure

10 Delivery Checklist

Item	File(s)	Status
RTL source (16 modules)	rtl/*.sv	✓
Directed testbench	tb/*.sv	✓
UVM testbench	uvm_tb/*.sv	✓
SVA assertion suites	sva/*.sv	✓
Formal FPV setups	formal/*.sby	✓
Synthesis script (DC)	scripts/synth_dc.tcl	✓
SDC timing constraints	constraints/*.sdc	✓
UPF power intent	rtl/*.upf	✓
IP-XACT descriptor	ip_xact/*.xml	✓
Coverage merge script	scripts/coverage_merge.sh	✓
Release package script	scripts/release_package.sh	✓
CDC waiver log	lint/cdc_waivers.md	✓
Integration guide	docs/integration_guide.md	✓
Specification	docs/latex/pcie_specification.pdf	✓
Project report	docs/latex/pcie_project_report.pdf	✓

Remaining for production signoff: FLIT-mode framing (v0.3.0), TLP ordering enforcement, UVM regression with commercial simulator coverage archive, CDC tool report review, synthesis timing closure at corner.

10 Acceptance Criteria

1. RTL compiles from a clean checkout using `filelists/rtl.f`.
2. `make sim` (iverilog) passes all 10 directed tests: 10/10 PASS.
3. UVM smoke and error regression tests complete without `UVM_FATAL` or `UVM_ERROR`.
4. Lint is either clean or all warnings are documented in `lint/waivers.vlt`.
5. All 5 CDC crossings are mitigated per `lint/cdc_waivers.md`.
6. Formal FPV (`formal/*.sby`) reports no assertion failures.
7. All known issues are documented in `docs/known_issues.md`.
8. Signoff checklist (`docs/signoff_checklist.md`) is fully reviewed before release.